RoboCore line sensor: 3x Vishay TCRT5000, analog outputs + LED indicators. JST SH 5 to RoboCore V2 J404.

Line sensor

3x TCRT5000, analog outputs
LM339 line indicator LEDs

File: line_sensor.kicad_sch

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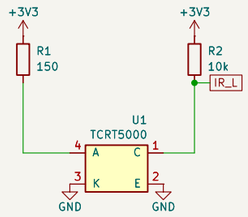
Sheet: /
File: RoboCore_LineSensor.kicad_sch

Title: RoboCore Line Sensor

Size: A4	Date: 2026-10-05	**Rev: 1.0**
KiCad E.D.A. 10.0.0		Id: 1/2

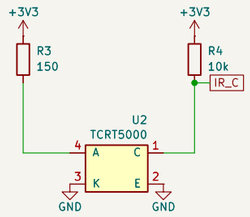
LEFT SENSOR

IR LED ~14 mA (150 from 3.3 V). Output IR_L: low over white, high over the line.



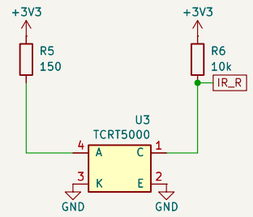
CENTRE SENSOR

IR LED ~14 mA (150 from 3.3 V). Output IR_C: low over white, high over the line.



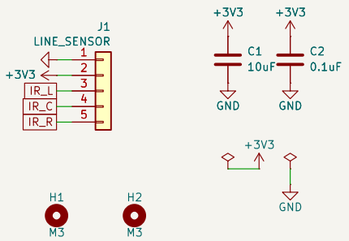
RIGHT SENSOR

IR LED ~14 mA (150 from 3.3 V). Output IR_R: low over white, high over the line.



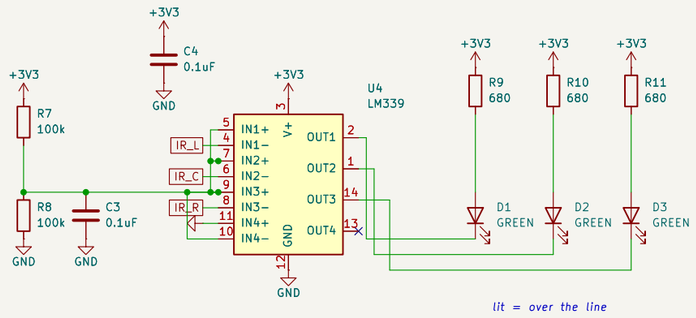
CONNECTOR

JST SH 5-pin to the RoboCore V2 (J404): GND, 3V3, L, C, R. Same pin order as the board's socket.



LINE INDICATORS (LM339)

Each comparator sinks its LED when the sensor voltage is above VREF (1.65 V), i.e. over the black line.
VREF = half of 3.3 V. The spare comparator has its inputs tied so its output stays low (pin 13 unused).



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Sheet: /Line sensor/
File: line_sensor.kicad_sch

Title: RoboCore Line Sensor – Line sensor

Size: A3	Date: 2026-10-05	**Rev: 1.0**
KiCad E.D.A. 10.0.0		Id: 2/2